

Space Propulsion - Formulary

Constants

	Gravity @ surface g_0 [m/s ²]	Mass M [kg]	Standard gravitational parameter μ [m ³ /s ²]	Mean radius R_0 [m]
Sun	274	1.989×10^{30}	1.327×10^{20}	6.957×10^8
Mercury	3.70	3.301×10^{23}	2.203×10^{13}	2.439×10^6
Venus	8.872	4.869×10^{24}	3.248×10^{14}	6.051×10^6
Earth	9.80665	5.971×10^{24}	3.986×10^{14}	6.378×10^6
Moon	1.62	7.346×10^{22}	4.904×10^{12}	1.737×10^6
Mars	3.72	0.642×10^{24}	4.282×10^{13}	3.389×10^6
Saturn	10.44	5.683×10^{26}	3.793×10^{16}	58.232×10^6
Uranus	8.87	8.681×10^{25}	5.793×10^{15}	25.362×10^6
Neptune	11.15	1.024×10^{26}	6.836×10^{15}	24.622×10^6

Universal gas constant:	$\Re = 8.314462 \frac{J}{mol K}$
Gravitational constant:	$G = 6.674 \times 10^{-11} \frac{m^3}{kg s^2}$
Astronomical unit AU:	$1 AU = 1.49598 \times 10^{11} m$
Solar constant at 1 AU:	$S_{\oplus} = 1366 \frac{W}{m^2}$
Elementary charge:	$e = 1.60217 \times 10^{-19} C$
Luminosity of Sun:	$L = 3.847 \times 10^{26} W$
Boltzmann constant:	$k = 1.38065 \times 10^{-23} \frac{J}{K}$
Velocity of light (vacuum):	$c = 2.997925 \times 10^8 \frac{m}{s}$

Electrical field constant: $\epsilon_0 = 8.85419 \times 10^{-12} \frac{As}{Vm}$

Magnetic field constant: $\mu_0 = 4\pi \times 10^{-7} \frac{Vs}{Am}$

Stefan-Boltzmann constant: $\sigma = 5.6704 \times 10^{-8} \frac{W}{m^2 K^4}$

Conversion: eV $\rightarrow$ J $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

Orbital mechanics

Table 1: Nomenclature

e	Eccentricity	[-]
a	Semi-major axis	[m]
b	Semi-minor axis	[m]
p	Semi-latus rectum	[m]
m, M	Mass	[kg]
r	Distance of centre of masses	[m]
r_{Peri}	Radius of shortest distance	[m]
r_{Apo}	Radius of largest distance	[m]
ϵ	Specific orbital energy	[km ² /s ²]
v_∞	Hyperbolic excess velocity	[m/s]
P	Orbital Period	[s]

Newton's gravitational law: $F_g = G \frac{m_1 m_2}{r^2}$

Gravitational acceleration: $g(r) = g_0 \frac{R_0^2}{r^2}$ for $r \geq R_0$ and $g_0 = \frac{\mu}{R_0^2}$

Energy law (Vis-Viva-Equation): $\frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} = \epsilon = \frac{v_\infty^2}{2} = \text{const.}$ while $\mu = GM$ & $c_3 = v_\infty^2$

Orbital velocity: $v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$

Satellite on circular orbit: $v_c^2 = \frac{\mu}{r_c}$

Orbital Geometry:

$$a = \frac{r_{Peri} + r_{Apo}}{2}$$

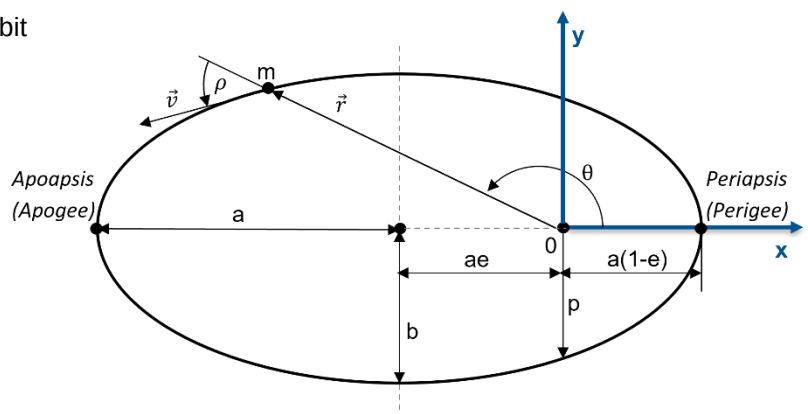
$$b = a\sqrt{1 - e^2}$$

$$p = a(1 - e^2)$$

$$r_{Peri} = a(1 - e) \quad \text{for elliptic \& hyperbolic orbit}$$

$$r_{Apo} = a(1 + e) \quad \text{for elliptic orbit}$$

$$r_{Peri} = \frac{p}{2} \quad \text{for parabolic orbit}$$



Conic section equations:

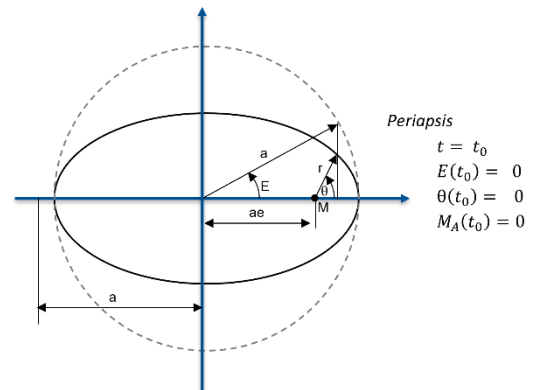
$$r = \frac{p}{1 + e \cos \theta} \quad (\text{ellipse, hyperbola})$$

$$1 = \frac{(x + ae)^2}{a^2} + \frac{y^2}{a^2(1 - e^2)} \quad (\text{ellipse, hyperbola})$$

$$y^2 = -2p \left(x - \frac{p}{2} \right) \quad (\text{parabola})$$

Time along Kepler trajectory:

θ :	True anomaly	$[-]$
E :	Eccentric anomaly	$[-]$
M_A :	Mean anomaly	$[-]$
t :	Time	$[s]$
M :	Central mass	$[kg]$
ε :	Specific trajectory energy	$\left[\frac{J}{kg} \right]$



Orbital Period: $P = \frac{2\pi a^{3/2}}{\sqrt{\mu}} = \frac{\pi \mu}{\sqrt{2}} (-\varepsilon)^{-3/2}$

Rendezvous:

$$\omega = \sqrt{\frac{\mu}{a^3}}$$

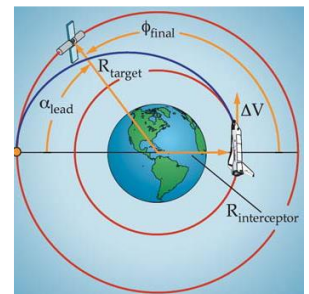
Objects angular velocity

$$TOF = \pi \sqrt{\frac{a_{transfer}^3}{\mu}}$$

Time of flight for half an orbit

$$\alpha_{lead} = \omega_{target} \cdot TOF$$

Lead angle



Time along an ellipse trajectory:

$$\cos \theta = \frac{\cos E - e}{1 - e \cos E}$$

Relationship $\theta \leftrightarrow E$

$$r = \frac{a(1 - e^2)}{1 + e \cos \theta}$$

Trajectory

$$r \cos \theta = a(\cos E - e)$$

Geometry

$$\frac{r \sin \theta}{a \sin E} = \sqrt{1 - e^2}$$

Geometry

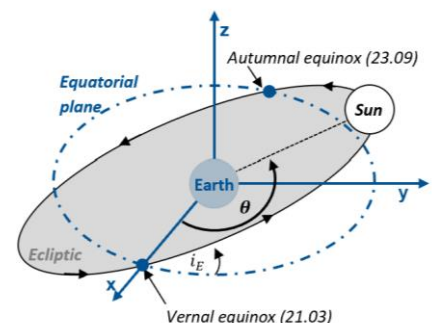
$$M_A = n(t - t_0) = E - e \sin E$$

Kepler's equation

$$n = \frac{2\pi}{T} = \sqrt{\frac{\mu}{a^3}} \quad \text{elliptic orbits}$$

while t_0 : time satellite passes periapsis

$$= \sqrt{-\frac{\mu}{a^3}} \quad \text{hyperbolic orbits}$$



Spacecraft Propulsion

Payload ratio: $\mu_{PL} = \frac{m_{PL}}{m_0}$

Structural ratio: $\mu_S = \frac{m_S + m_M}{m_0}$

Propellant ratio: $\mu_{PR} = \frac{m_{PR}}{m_0}$

Tsiolkovsky: $\Delta v = c_e \cdot \ln\left(\frac{m_0}{m_f}\right)$

$$\Delta v = I_{sp} \cdot g_{0,earth} \cdot \ln\left(\frac{m_0}{m_f}\right)$$

$$\Delta v = c_e \cdot \ln\left(\frac{1}{\mu_S + \mu_{PL}}\right)$$

Sequential staging: $\Delta v_{total} = \Delta v_1 + \Delta v_2 + \dots = c_{e,1} \ln \frac{m_{0,1}}{m_{f,1}} + c_{e,2} \ln \frac{m_{0,2}}{m_{f,2}} + \dots$

$$\frac{\Delta v}{c_e} = \sum_{i=1}^N \frac{\Delta v_i}{c_e} = \sum_{i=1}^N \ln\left(\frac{1}{1 - \mu_{PR_i}}\right)$$

Propulsion Demand

Inclination adjustment: $\Delta v_{inc} = \sqrt{v_1^2 + v_2^2 - 2v_1v_2 \cos \Delta i}$

Ascent velocity: $v_{ascent} = \sqrt{2 \cdot g_0 \left(R_0 - \frac{R_0^2}{R_0 + h}\right)}$

Ascent into circular orbit: $\Delta v_{ideal} = \sqrt{v_{ascent}^2 + v_{orbit}^2} = \sqrt{g_0 R_0 \left(2 - \frac{R_0}{R_0 + h}\right)}$

Gravity Losses: $\Delta v_{gravity} = \sqrt{2 c_e g t \sin(\theta) \ln\left(\frac{m_0}{m_f}\right) + g^2 t^2}$ with θ = launching-angle

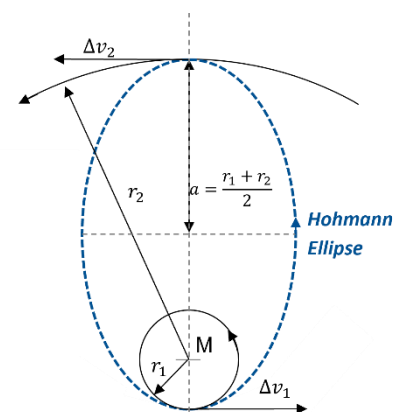
Hohmann Transfer: $\Delta v_{HO} = \Delta v_1 + \Delta v_2$

Spiraling Transfer:

- Continuous, weak thrust (Thrust $F \ll$ Gravitational Force $\frac{\gamma m M}{r^2}$)
- Thrust direction ($\vec{F}$) tangential to the orbit direction ($\vec{s}_0$),
at all times it is valid: $\vec{v} \uparrow \vec{F}$
- Trajectory "circle-like" at all times

Spiralling up from r_0 to r : $\Delta v_{ST} = v_{ST,0} - v_{ST} = \sqrt{\mu} \left(\sqrt{\frac{1}{r_0}} - \sqrt{\frac{1}{r}} \right)$ with

Trajectory:
$$r = \frac{r_0}{\left\{ 1 + \sqrt{\frac{r_0 F}{\mu m}} \ln \left[1 - \frac{m}{m_0} (t - t_0) \right] \right\}^2}$$



Idealised Thermal Engine

Function Γ :
$$\Gamma = \sqrt{\kappa \left(\frac{2}{\kappa+1} \right)^{\frac{\kappa+1}{\kappa-1}}}$$

Thrust:
$$F = \dot{m} \cdot v_e + (p_e - p_a)A_e$$

$$F = \dot{m} \cdot c_e$$

$$F = p_0 A_t \left[\Gamma \sqrt{\frac{2\kappa}{\kappa-1}} \sqrt{1 - \left(\frac{p_e}{p_c} \right)^{\frac{\kappa-1}{\kappa}}} + \frac{A_e}{A_t} \left(\frac{p_e - p_a}{p_c} \right) \right]$$

Oxidiser-to-fuel ratio:
$$ROF = \frac{\dot{m}_{ox}}{\dot{m}_{fu}}$$

Effective velocity:
$$c_e = \frac{F}{\dot{m}} = c^* \cdot c_F$$

Specific impulse:
$$I_{sp} = c_e / g_{0,earth} \quad \text{while} \quad I_{sp,total} = \frac{1}{\dot{m}_{total}} \sum_{i=1}^n \dot{m}_i I_{sp,i}$$

Exhaust velocity:
$$v_e = \sqrt{2 \frac{\kappa}{\kappa-1} \cdot \frac{\mathfrak{R}}{M} \cdot T_c \cdot \left(1 - \left(\frac{p_e}{p_c} \right)^{\frac{\kappa-1}{\kappa}} \right)}$$

Characteristic velocity:
$$c^* = \frac{p_c \cdot A_{th}}{\dot{m}}; c_{theo}^* = \sqrt{\frac{\mathfrak{R} \cdot T_c}{M \cdot \kappa} \left(\frac{\kappa+1}{2} \right)^{\frac{\kappa+1}{\kappa-1}}}$$

Thrust coefficient:
$$c_F = \frac{F}{p_c A_{th}} = \frac{\dot{m} v_e}{p_c A_{th}} + \varepsilon \left(\frac{p_e - p_a}{p_c} \right)$$

Summerfield-criterion:
$$\frac{p_e}{p_a} \approx 0,25 - 0,4$$

Expansion area ratio:
$$\varepsilon = \frac{A_e}{A_{th}} = \Gamma \left(\frac{p_e}{p_c} \right)^{-\frac{1}{\kappa}} \left\{ \frac{2\kappa}{\kappa-1} \left[1 - \left(\frac{p_e}{p_c} \right)^{\frac{\kappa-1}{\kappa}} \right] \right\}^{-\frac{1}{2}}$$

Tanks / pressure vessels

Barlow's formula (spherical):
$$t_{min,sphere} = \frac{pd}{4\sigma_{max}} * SF$$

Barlow's formula (cylindrical):
$$t_{min,cylinder} = \frac{pd}{2\sigma_{max}} * SF$$

Approx. mass spherical tank:
$$m_{Ta} = 4 \cdot \pi \cdot r_i^2 \cdot t \cdot \rho_{Ta}$$

Pressurant volume (polytropic state change):
$$V_1 = V_3 \cdot \frac{p_3 \cdot T_2}{p_2 \cdot T_3} \left[\left(\frac{p_1}{p_2} \right)^{\frac{1}{n}} - 1 \right]^{-1}$$

Thrust chamber assembly

Characteristic length:
$$l^* := \dot{m} \frac{\Delta t_c}{\rho_c \cdot A_{th}} = \frac{V_c}{A_{th}}$$

Expansion ratio:
$$\varepsilon := \frac{A_e}{A_{th}}$$

Turbopumps

Pump head: $p_P = p_c + \Delta p_{inj,CC} + \Delta p_{cool} + \Delta p_{L\&V} + \Delta p_T + \Delta p_{inj,PB} - p_s$

Loss cooling channels: $\frac{\Delta p_{cool}}{p_c} = k \cdot \frac{p_c^n}{F^m}; p_c[bar], F[kN]$

Required pump power: $P_P = \frac{\dot{V}}{\eta_P} \cdot \Delta p = \frac{(\frac{\dot{m}}{\rho})}{\eta_P} \cdot \Delta p$

Power balance turbine: $P_T = \eta_T \cdot \dot{m}_T \cdot c_p \cdot (T_{in} - T_e) = \eta_T \cdot \dot{m}_T \cdot c_p \cdot T_{in} \cdot \left(1 - \left(\frac{p_e}{p_{in}}\right)^{\frac{\kappa-1}{\kappa}}\right)$

Mass flow fraction: $k_A = \frac{\dot{m}_T}{\dot{m}_{ox+fu}} = \frac{\Delta p_{P,F}}{\rho_F} \cdot \frac{1}{\eta_{TP}} \cdot \frac{1}{1+OF} \cdot \frac{1}{T_{GG} \cdot c_p \left[1 - \left(\frac{p_e}{p_{GG}}\right)^{\frac{\kappa-1}{\kappa}}\right]}$

Combined efficiency: $\frac{1}{\eta_{TP}} = \frac{1}{\eta_{P, fu} \cdot \eta_{T, fu}} + \frac{\Delta p_{P, ox}}{\Delta p_{P, fu}} \cdot \frac{\rho_{fu}}{\rho_{ox}} \cdot OF \cdot \frac{1}{\eta_{P, ox} \cdot \eta_{T, ox}}$

Turbine pressure ratio: $\Pi_T := \frac{p_{in}}{p_e}$

Cooling

1D-Conduction: $\dot{q}_{cond} \left[\frac{W}{m^2} \right] = \lambda \frac{T_{wg} - T_{wc}}{t_{hg}}$

1D-Convection: $\dot{q}_{conv} \left[\frac{W}{m^2} \right] = \alpha_{wg}(T_{hg} - T_{wg}) = \alpha_{wc}(T_{wc} - T_{cool})$

Nusselt number: $Nu = \frac{\alpha L}{\lambda}$

Cinjarew correlation: $Nu = 0,0162 \cdot (Re \cdot Pr)^{0,82} \left(\frac{T_{hg}}{T_{wg}}\right)^{0,35}$

Reynolds number: $Re = \frac{\rho v L}{\mu}$

Prandtl number: $Pr = \frac{\mu c_p}{\lambda}$

Hydraulic diameter: $d_h = 4 \frac{A}{U}$

Solid Propulsion

Burning rate: $\dot{r} = a \cdot p_c^n$

Mass flow balance (steady): $\dot{m} = \rho_{PR} \cdot A_B(t) \cdot \dot{r} = \frac{p_c \cdot A_{th}}{c^*}$

Attitude Determination and Control

Stabilization of angular momentum

General relationships (Assumption $\Theta_x = \Theta_y$):

$$\omega_q = \sqrt{\omega_x^2 + \omega_y^2}$$

$$\tan \vartheta_N = \frac{\Theta_x \omega_q}{\Theta_z \omega_z} = \frac{\omega_q}{\omega_N}$$

$$\omega_N = \frac{\Theta_z \omega_z}{\Theta_x}$$

Passive nutation damping: $\vartheta_N(t) = \vartheta_N(t_0) e^{\left(-\frac{t}{T_D}\right)}$ (condition: $\Theta_z > \Theta_x, \Theta_y$)

$\vartheta_N(t)$: Half-opening angle of nutation cone [deg]

$\vartheta_N(t_0)$: Start value of ϑ_N [deg]

t: Time [s]

T_D : Damping constant [s]

Θ : Moment of inertia [kgm²]

ω : Angular frequency $\left[\frac{1}{s}\right]$

Thermodynamics

Polytropic process: $\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$

Isentropic exponent (ideal gas): $\kappa = c_p / c_v$

Isentropic process: $\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} = \left(\frac{\rho_2}{\rho_1}\right)^{\kappa-1}$ or $\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^\kappa$

Ideal gas law: $pV = mRT$

Continuity: $\dot{m} = \rho v A$

Energy System

Sun-Angle (Beta-angle):

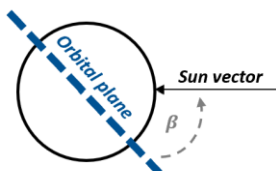
$$\sin \beta = \cos \theta \sin \Omega \sin i - \sin \theta \cos i_E \cos \Omega \sin i + \sin \theta \sin i_E \cos i$$

$i_E = 23.5^\circ$, Inclination of earth's equatorial plane with ecliptic

Ω = Right ascension of ascending node of orbit

i = Inclination of orbit

θ = Right ascension of sun



Eclipse Time:

$$t_e = \frac{p}{\pi} \arccos \frac{\sqrt{r^2 - R^2}}{r \cos \beta}$$

Geometry

Sphere: $V_{sphere} = \frac{4}{3}\pi r^3$ $S_{sphere} = 4\pi r^2$

Cylinder: $V_{cylinder} = \pi r^2 h$ $S_{cylinder} = 2\pi r h + 2\pi r^2$

Circle: $A_{circle} = \pi r^2$

Moment of inertia rectangle: $I_{xx} = \frac{1}{12}m(a^2 + b^2)$

Satellite Design

Battery Sizing

Minimum capacity $C_{min} = P_{req_e} t_e$

Battery Capacity $C_{battery} = \frac{C_{min}}{DoD}$

C_{min} : Capacity needed for eclipse [Wh]

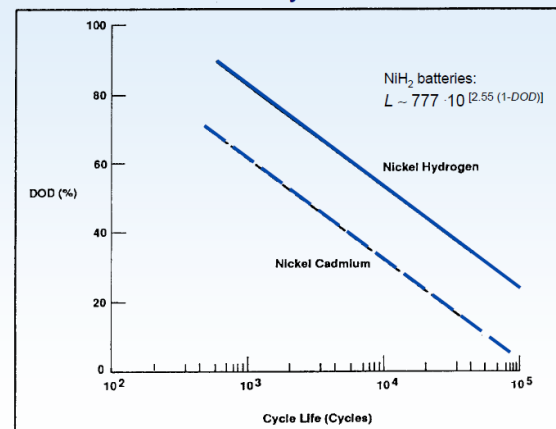
P_{req_e} : Required power during eclipse [W]

$C_{battery}$: Capacity of battery [Wh]

t_e : Time of eclipse [s]

DoD : Depth of discharge [%]

Depth-of-Discharge vs. Cycle Life for Secondary Batteries



SMAD, Fig. 11.11

Solar panel sizing

Solar Array Power Output:

$$P = \phi_{sun} \eta_{array} A_{array}$$

P : Power output [W]

ϕ_{sun} : Flux of sunlight at the orbit (in earth orbit: $\phi_{sun} = 1366 \left[\frac{W}{m^2}\right]$)

η_{array} : Conversion efficiency at operating temperature [-]

A : Area of solar array [m^2]

Thermal Subsystem Design

Sun radiation: $Q_s = \alpha \phi_{sun} A_{sun \perp sat}$

Albedo radiation: $Q_a = \alpha \phi_{albedo} A_{earth \perp sat} F$

Planetary (earth) radiation: $Q_{earth} = \alpha q_I A_{earth \perp sat}$

Emitted radiation satellite: $Q_{emitted} = \varepsilon \sigma A_{sat} T^4$

q_I : Avg. IR energy flux from earth $237 \pm 21 \frac{W}{m^2}$

α Absorptivity [-]

ε Emissivity [-]

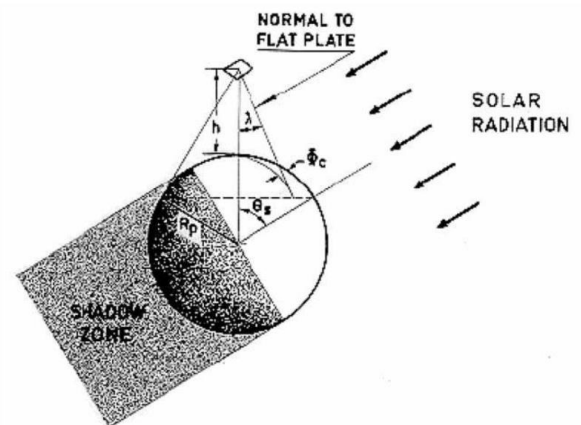
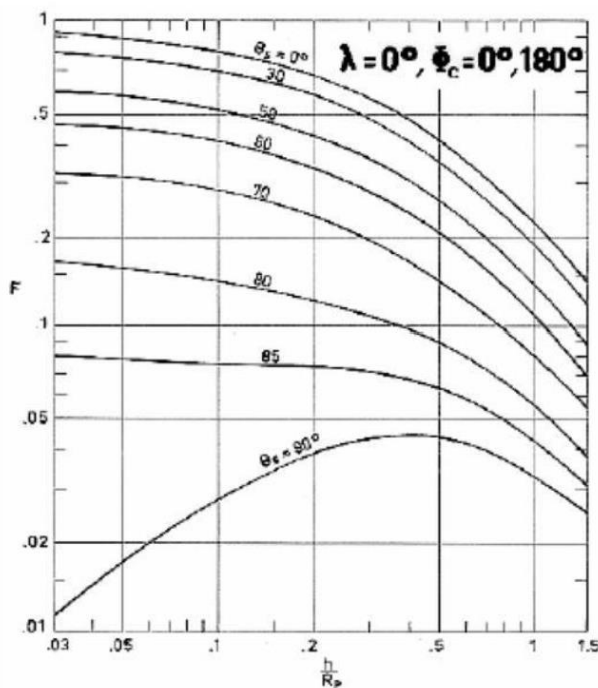
ϕ_{sun} : Flux of sunlight at the orbit (in earth orbit: $\phi_{sun} = 1366 \frac{W}{m^2}$)

ϕ_{albedo} : Flux of the Sun reflected on the Earth surface (around 37% of ϕ_{sun})

F: View factor [-]

T: Skin temperature of satellite [K]

σ : Stefan-Boltzmann constant $\sigma = 5.6704 \times 10^{-8} \frac{W}{m^2 K^4}$



Optical Sizing

Ground Resolution Distance $GRD = 2.44 \frac{h\lambda}{D}$

GRD: Ground resolution distance [m]

h: Altitude [m]

D: Circular aperture of diameter [m]

lambda: Averaged wavelength [m]

TT&C Subsystem

Wavelength: $\lambda = \frac{c}{f}$

Received power in linear scale $P_{RX} = \frac{P_{TX} G_{TX} G_{RX}}{L_{TX} L_{FS} L_{RX}}$

Free space losses in linear scale $L_{FS} = \left(\frac{4\pi d f}{c} \right)^2$

Noise power is $N_{RX} = k(T_A + T_R)B$ $(k = 1.38065 \times 10^{-23} \frac{J}{K})$

Signal to noise ratio in linear scale $SNR = \frac{P_{RX}}{N_{RX}}$

Shannon-Hartley -Theorem: $C = B \cdot \log_2 \left(1 + \frac{S}{N} \right)$

$C = B \frac{SNR}{3,01}$

C : Channel Capacity (bit/s)

B : Bandwidth of the channel (Hz)

S : Average Signal Power

N : Average Noise Power

d : Distance [m]

f : Frequency [Hz]

c : Speed of light [m/s]

P : Power

G : Gain

L : Loss

RX : Receiving

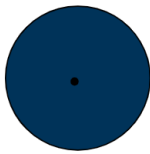
TX : Transmitting

Antenna Gain

Parabolic Antenna:

$$G_t = \eta \left(\frac{D \cdot \pi}{\lambda} \right)^2 = \eta \left(\frac{4 \cdot \pi}{\lambda} \right)^2 A_t$$

Antenna efficiency η : typ. 0.2 (WiFi, obstructions from feed, shape, etc.) to < 1.0



No Gain ($G=1$) -
Isotropic



Low Gain =
Broad Beam



High Gain =
Narrow Beam

Reaction Wheels (RW) sizing

Angular velocity of reaction wheel: $\omega_{RW} = \frac{L_{RW}}{M_{MAXRW}} \omega_{MAXRW}$

Time manoeuvre with RW: $t_{manoeuvre} = \frac{\Delta L_{SAT}}{M_{RW}}$

L_{RW} :	Angular momentum RW
M_{MAXRW} :	Maximum momentum that the RW can provide
ω_{MAXRW} :	Maximum angular velocity of the RW
L_{SAT} :	Angular momentum satellite

Magnetorquer sizing

Magnetic dipole: $\vec{m} = NI\vec{A}$

With:

- N : number of windings
- I : Current [A]
- A : Area [m²]

Magnetic field of the Earth at 500 km over Munich: $B_E = 3.9 \cdot 10^{-5} \text{ T}$

Torque: $\vec{T} = \vec{m} \times \vec{B}_E$

Indices / Nomenclature

O: Initial

a: Ambient

A: Area

B: burning surface

C: (Combustion) Chamber

cool: Coolant / Cooling channels

e: Exit plane

eff: Effective

f: final

fu: Fuel

GG: Gas generator

hg: Hot gas

HO: Hohmann

in: Inlet

inc: Inclination

inj: Injection

L&V: Lines & valves

M: Motor

N: Nutation

Ox: Oxidiser

P: Pump

PB: PreBurner

PL: Payload

PR: Propellant

S: Structure / Surface

SF: Safety Factor

sp: Specific

ST: Spiraling Transfer

T: Turbine

Ta: Tank

th: Throat

theo: Theoretical

U: Circumference

V: Volume

wc: Wall on coolant side

wg: Wall on hot gas side